

PROFESSIONAL PROFILE:

- Bring 10+ years of hands-on experience in information technology specializing in Big Data Engineering, delivering impactful data solutions across healthcare, retail, banking, logistics, and manufacturing sectors. Demonstrated expertise in building enterprise-scale data platforms that transform raw information into actionable business intelligence for diverse stakeholders.
- Engineered distributed ingestion frameworks to centralize multi-channel transactional feeds into **Azure Data Lake Storage Gen2** using **Apache NiFi**, mapping schema hierarchies inside **Erwin** configurations to eliminate architectural silos and accelerate downstream analytics across cloud business intelligence applications.
- Formulated real-time tracking loops by funneling operational event telemetry through **Azure Event Hubs**, executing transformation policies via **Azure Stream Analytics** engines, and writing the cleansed records back to high-throughput persistent storage clusters without pipeline latency.
- Constructed heavy data manipulation pipelines using **Azure Data Factory**, reshaping complex multi-source vendor lists via **dbt** scripts, and systematically loading the audited relational tables directly into target layers across **Azure Synapse Analytics**.
- Modeled low-latency user profiles inside distributed **Azure Cosmos DB** instances, connecting back-end services engineered with **Java** to fetch catalog inventories securely while enforcing validation standards through custom cloud **OAuth2** authentication gateways.
- Deployed containerized analytical microservices onto managed **Azure Kubernetes Service (AKS)** clusters, standardizing infrastructure updates via **Git** repository structures and automating daily continuous integration steps through centralized enterprise **Jenkins** build loops.
- Provisioned isolated multi-region compute environments using **Terraform** orchestration templates, setting up sovereign host layers on **Azure Virtual Machines** while mapping resource governance boundaries through explicit **Azure Resource Manager** blueprints.
- Automated document indexing loops by processing unorganized physical supply chain invoices inside **Azure Form Recognizer**, validating text structures via **Python**, and compiling clean outputs into enterprise search indices using **Azure Cognitive Search**.
- Trained advanced item recommendation classifiers within **Azure Machine Learning** instances, using historical transaction sheets aggregated via **Pandas** to isolate shifting demand spikes during high-volume commercial shopping periods.
- Initiated serverless backend computation logic using **Azure Functions** to ingest real-time digital interaction logs, passing raw strings through **Azure Cognitive Services** text models, and persisting the resulting insights inside partitioned **Parquet** formats.
- Constructed real-time message brokering clusters using the **Confluent Platform**, consuming incoming clickstream logs via **Kafka Streams** components, and systematically synchronizing the finalized transactional summaries into secure cloud **Delta Lake** storage.
- Established extensive automated data quality check modules across unified **Microsoft Fabric** workspaces, verifying underlying pipeline integrity metrics via **Great Expectations** assertions to guarantee schema alignment across modern **Apache Iceberg** tables.
- Maintained end-to-end platform observability across global microservices by monitoring container clusters with **Prometheus**, analyzing application telemetry via **Azure Application Insights**, and constructing technical operational status boards using **Grafana** displays.
- Formulated conversational customer support assistants using **Azure Bot Services**, deploying foundational semantic models via **Azure OpenAI Service**, and tailoring operational performance metrics inside collaborative spaces guided by corporate **DevOps** methodologies.
- Structured detailed asset mapping frameworks by establishing comprehensive dictionary layers within **Collibra**, integrating asset discovery via **Azure Data Catalog**, and monitoring transactional reliability indicators across multi-cloud integrations through automated **Monte Carlo** alerts.

- Evaluated historical consumer trend matrices within **Azure Databricks** compute workspaces, running parallel array manipulations via **NumPy** libraries, and highlighting significant regional revenue variations through interactive dashboards built inside **Power BI**.
- Monitored serverless cluster workloads by capturing system exceptions with **Azure Monitor**, isolating unexpected pipeline drop-offs within **Azure Log Analytics**, and scaling background execution worker groups via automated **Azure Batch** triggers.
- Developed complex analytical matrix transformations inside **Azure AI Studio**, training deep neural networks using the **PyTorch** library, and creating granular spatial monitoring dashboards to analyze hardware performance via **Matplotlib** packages.
- Engineered distributed stream processing architectures on **Google Kubernetes Engine (GKE)** spaces, ingesting real-time message streams with **Cloud Dataflow**, and storing continuous financial data safely within cloud-managed **Google Cloud Storage** buckets.
- Standardized enterprise database migration pipelines via **Data Fusion**, extracting historical accounting records out of old **Hadoop** directory structures, and loading the resulting records directly into partitioned enterprise **BigQuery** targets.
- Built predictive financial risk classification loops inside **Vertex AI** clusters, optimizing hyperparameter training spaces via **Scikit-learn** algorithms, and displaying model performance drift timelines using specialized analytical scripts in **Seaborn**.
- Orchestrated automated batch computation flows by defining complex directed acyclic graphs inside **Cloud Composer**, consolidating structural inventory records across **HBase** layers, and cleaning messy data fields using **Spark SQL** queries.
- Managed secure data sharing pathways within **GCP** nodes, safeguarding persistent database fields using **Cloud Key Management Service** layers to enforce **Encryption at Rest** while isolating cluster access permissions via **Cloud IAM** policies.
- Formulated robust automated model training frameworks within **AI Platform (legacy)** configurations, leveraging predictive algorithms built on **TensorFlow** libraries, and automating continuous integration updates using integrated enterprise **GitHub Actions** workflows.
- Implemented interactive automated response systems by connecting **Dialogflow** interfaces with **Speech-to-Text** audio translators, validating incoming parameters with **Cloud Functions**, and outputting immediate synthesis confirmations using cloud-native **Text-to-Speech** blocks.
- Maintained comprehensive operational logging visibility across distributed trading environments by aggregating cloud metadata histories into **Cloud Logging**, tracking system health thresholds inside **Cloud Monitoring**, and optimizing resource utilization parameters under **Agile** frameworks.
- Curated massive statistical customer indexes using **AutoML** modeling options, applying linguistic analysis parameters via **Cloud Vision API** nodes, and documenting pipeline changes within shared project workspaces governed under structured **Scrum** guidelines.
- Consolidated enterprise metadata governance structures within **Dataplex** spaces, evaluating machine learning models built on **Keras** using **BigQuery ML** functionalities, and visualizing backend performance trends via an enterprise-scale **ELK Stack**.

TECHNICAL SKILLS

- **Big Data Ecosystem:** Apache Spark, PySpark, Spark SQL, Spark Streaming, Apache Flink, Hadoop, HDFS, YARN, MapReduce, Hive, HBase, Apache Kafka, Kafka Streams, Kafka Connect, Confluent Platform, Apache Airflow, Apache NiFi, Apache Flume, Sqoop, Zookeeper, Delta Lake, Apache Iceberg, Parquet, ORC
- **ETL / ELT Tools:** Azure Data Factory (ADF), AWS Glue, Google Cloud Dataflow, Informatica PowerCenter, Talend Open Studio, SSIS, Semarchy xDM, dbt, Apache Beam, DataStage, Data Fusion, AWS Data Pipeline, Azure Stream Analytics
- **Programming Languages:** Python, PySpark, Scala, Java, SQL, T-SQL, Bash, Unix Shell Scripting, Spring Boot, NodeJS, REST APIs, OAuth2
- **Cloud Environment:** AWS (EMR, S3, Glue, Redshift, Lambda, Athena, EC2, RDS, DynamoDB, Kinesis, CloudWatch, CloudTrail, ECS, IAM, VPC, SNS/SQS, CloudFormation), Azure (Data Factory, Databricks, Synapse Analytics, Data Lake Storage Gen1/Gen2, Blob Storage, SQL Database, Cosmos DB, HDInsight, Event Hubs, Stream Analytics,

Functions, AKS, Batch, Monitor, Log Analytics, Application Insights, ARM, Virtual Network, Azure AD, Azure DevOps, Microsoft Fabric), GCP (BigQuery, Dataflow, Dataproc, Cloud Storage, GKE, Cloud Composer, Pub/Sub, Cloud Functions, Compute Engine, Vertex AI, Dataplex, Dialogflow, Cloud Monitoring, Cloud Logging, Cloud IAM, Cloud KMS), Snowflake

- **AI / Machine Learning:** Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, Hugging Face Transformers, NumPy, Pandas, Matplotlib, Seaborn, Random Forest, Logistic Regression, Decision Trees, Gradient Boosting, Azure Machine Learning, Azure OpenAI Service, Vertex AI
 - **Databases & Data Warehousing:** Oracle 11g/10g, SQL Server 2014/2012, MySQL, PostgreSQL, Teradata 15/14, BigQuery, Amazon Redshift, Azure Synapse Analytics, Snowflake, Cosmos DB, MongoDB, DynamoDB, HBase, Star Schema, Snowflake Schema, Facts & Dimensions Modeling
 - **Containerization & DevOps Tools:** Docker, Kubernetes, AKS, GKE, ECS, Jenkins, GitHub Actions, GitLab CI, Azure DevOps, Terraform, ARM Templates, CloudFormation, Git, SVN, CI/CD Pipelines
 - **Monitoring & Observability Tools:** Grafana, Prometheus, Splunk, ELK Stack, Azure Monitor, Azure Log Analytics, Azure Application Insights, AWS CloudWatch, Cloud Monitoring, Cloud Logging, Great Expectations, Monte Carlo
 - **Reporting & Visualization Tools:** Power BI, Tableau, Looker, Amazon QuickSight, Grafana, Excel
 - **Data Governance & Modeling Tools:** Collibra, Alation, Dataplex, Azure Data Catalog, Apache Atlas, ER/Studio, Erwin, PowerDesigner
 - **Security & Compliance:** IAM, Azure Active Directory, RBAC, Encryption at Rest, Encryption in Transit, HIPAA, PCI-DSS, Secure API Design, Network Security, Private Endpoints
 - **Methodologies:** Agile, Scrum, SDLC, Waterfall, DevOps, Sprint Planning, Code Reviews, Production Support
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PROFESSIONAL EXPERIENCE

Nike | Beaverton, OR

Sr. Data Engineer | March 2025 – Current

- Developed automated ingestion pipelines to centralize multi-channel retail transactions into **Azure Data Lake Storage Gen2** using **Apache NiFi**, establishing structured historical schemas with **Erwin** data models, which successfully reduced initial data preparation latency by 35% across downstream financial reporting applications.
- Constructed a real-time inventory tracking framework utilizing **Apache Kafka** to stream point-of-sale events, applying complex transformations via **PySpark** to maintain accurate stock levels across regional distribution hubs before persisting the refined transactional datasets directly into **Delta Lake** storage.
- Orchestrated daily promotional data extractions using **Azure Data Factory**, executing multi-stage transformations with **dbt** to sanitize unstructured retail vendor inputs, validating complete schema compliance via **Great Expectations**, and ultimately updating centralized warehouse tables within **Azure Synapse Analytics**.
- Designed a predictive product demand system by training custom **XGBoost** regression architectures within **Azure Machine Learning**, leveraging historical sales figures cleaned via **Pandas** to provide granular stock requirements for individual storefronts during peak holiday shopping seasons.
- Deployed containerized microservices for customer order tracking onto **Azure Kubernetes Service (AKS)**, leveraging declarative configuration files managed through **Git** version control and automating continuous deployment pipelines via **Jenkins** to ensure seamless production application updates.
- Modeled complex e-commerce user profiles within **Azure Cosmos DB**, leveraging native distributed indexing alongside **Java** backend microservices to fetch personalized retail catalog data under low-latency constraints while using **OAuth2** validation protocols to securely protect customer profiles.
- Provisioned multi-region cloud environments utilizing **Terraform** infrastructure-as-code scripts, configuring isolated subnets on **Azure Virtual Machines** to host specialized processing nodes while systematically managing cloud resources via **Azure Resource Manager** blueprints under a structured **DevOps** framework.

- Implemented automated customer sentiment processing pipelines using **Azure Functions** to ingest online product reviews, extracting core semantic meaning via **Azure Cognitive Services** before structuring the finalized textual insights inside high-performance columnar **Parquet** storage formats.
- Conducted extensive customer churn analysis by processing millions of loyalty member interaction records using **Apache Spark**, utilizing optimized **Spark SQL** queries to identify high-risk accounts and displaying the resultant retention trends via interactive **Power BI** executive dashboards.
- Built an automated inventory audit pipeline by orchestrating high-throughput data migrations with **Apache Airflow**, converting nested JSON web payloads into structured rows through **Python** scripts, and storing the organized outputs securely within **Azure SQL Database** instances.
- fine-tuned deep learning classification models utilizing **TensorFlow** to automatically categorize incoming product return images, leveraging distributed GPU compute clusters inside **Azure AI Studio** to significantly accelerate model training times for the quality assurance team.
- Established an observability framework for retail checkout systems by collecting telemetry through **Azure Monitor**, analyzing performance anomalies via **Azure Log Analytics**, and visualizing transaction drop-off points using custom **Grafana** dashboards to ensure high availability.
- Formulated real-time surge pricing engines by utilizing **Apache Flink** to process continuous streaming competitor web metrics, enriching data with internal retail margins using **Scala**, and publishing immediate price adjustments directly back to **Confluent Platform** clusters.
- Constructed an internal supply chain knowledge base by parsing supplier invoices with **Azure Form Recognizer**, extracting key payment entities, and indexing the unstructured text within **Azure Cognitive Search** to allow procurement teams to query historical procurement documents.
- Executed heavy batch processing jobs for seasonal price optimization across extensive catalogs using **Azure Batch**, applying mathematical matrix operations via **NumPy** to evaluate margin elasticity before pushing updated retail price lists into production databases.
- Monitored distributed processing health across streaming clusters by scraping systems metrics with **Prometheus**, tracing microservice execution delays using **Azure Application Insights**, and coordinating technical triage workflows under structured **Agile** sprint parameters to maintain platform reliability.
- Developed an interactive product recommendation chatbot using **Azure Bot Services** to assist online shoppers, integrating foundational large language models via **Azure OpenAI Service** to deliver context-aware answers and personalized retail promotion details based on browsing intent.
- Created an automated data governance catalog using **Collibra** to document critical retail assets, utilizing **Azure Data Catalog** to map lineage across pipelines and employing **Monte Carlo** to alert data teams of unexpected distribution anomalies.
- Engineered a real-time clickstream analytics platform by funneling online retail traffic through **Azure Event Hubs**, utilizing **Azure Stream Analytics** to compute rolling conversion metrics, and routing immediate telemetry summaries into enterprise-wide **Microsoft Fabric** workspaces.
- Standardized internal machine learning development lifecycles by packaging custom **PyTorch** forecasting scripts inside isolated **Docker** containers, which were then smoothly executed across on-demand compute instances using **Azure Container Instances** to lower recurring infrastructure overhead.
- Analyzed regional market basket data by applying **Scikit-learn** association rules to historical customer receipts, utilizing **Seaborn** to map product purchase correlations and organizing the resultant statistical distributions inside comprehensive **Matplotlib** visualizations for merchandising teams.
- Automated text translation workflows for international e-commerce listings using **Hugging Face Transformers**, deploying semantic translation models via **Keras** libraries, and leveraging **Azure Speech Services** to generate accessible audio descriptions for global mobile applications.
- Optimized massive product dimension tables by converting legacy flat files into optimized **Apache Iceberg** table formats, utilizing cloud storage solutions within **Azure Blob Storage** to guarantee transaction consistency during concurrent data pipeline updates.
- Managed secure corporate access controls across retail administrative portals by configuring enterprise identity directories within **Active Directory**, executing authorization policies via custom **Bash** operational scripts to strictly enforce least-privilege security paradigms across global departments.
- Trained advanced item classification systems using **LightGBM** on historical promotional responses, processing localized customer datasets via **Kafka Streams**, and deploying the trained models directly onto high-throughput cloud clusters during synchronized **Scrum** product release cycles.

- Created comprehensive visual defect inspection pipelines for distribution centers using **Azure Vision Services**, processing high-resolution conveyor belt feeds with custom **SQL** lookup parameters to automatically isolate damaged merchandise packaging before final shipping.
- Streamlined enterprise reporting structures by unifying disparate transactional layers into **Microsoft Fabric**, executing complex analytical transformations using **Spark Streaming** models, and securing data assets across all cloud boundaries with robust **OAuth2** authentication protocols.

Tech Stack: Azure Virtual Machines, Azure Blob Storage, Azure Kubernetes Service (AKS), Azure Container Instances, Active Directory, Azure Monitor, Azure Log Analytics, Azure Application Insights, Azure DevOps, Azure Resource Manager, Azure Functions, Azure Batch, Azure OpenAI Service, Azure Machine Learning, Azure Cognitive Services, Azure AI Studio, Azure Bot Services, Azure Cognitive Search, Azure Form Recognizer, Azure Speech Services, Azure Vision Services, PyTorch, Scikit-learn, TensorFlow, Keras, XGBoost, LightGBM, Hugging Face Transformers, NumPy, Pandas, Matplotlib, Seaborn, Azure Data Factory, Azure Synapse Analytics, Azure Data Lake Storage Gen2, Azure SQL Database, Azure Cosmos DB, Azure Databricks, Azure Stream Analytics, Azure Event Hubs, Azure Data Catalog, Microsoft Fabric, Apache Spark, PySpark, Spark SQL, Spark Streaming, Apache Flink, Delta Lake, Apache Iceberg, Parquet, Apache Kafka, Kafka Streams, Confluent Platform, Apache NiFi, Apache Airflow, dbt, Erwin, Collibra, Great Expectations, Monte Carlo, Power BI, Prometheus, Grafana, Git, Jenkins, Terraform, Docker, Kubernetes, Python, Java, Scala, SQL, Bash, OAuth2, Agile, Scrum, DevOps

Huntington | Columbus, OH

Sr. Data Engineer | May 2023 – February 2025

- Engineered a scalable loan underwriting pipeline by capturing core core-banking transactional logs with **Apache NiFi**, transferring the raw events into structured **Parquet** storage formats, and orchestrating downstream ledger balancing tasks seamlessly through dedicated **Cloud Composer** DAG workflows.
- Formulated secure transaction analysis systems using **Cloud Dataflow** to ingest payment messages from legacy sources via **Kafka Connect**, decrypting incoming payloads in real time before mapping customer spending records directly into target schemas inside **BigQuery**.
- Established distributed batch frameworks on **Cloud Dataproc** to process decades of historical mortgage documentation, executing optimized **PySpark** routines to clean messy fields and persisting organized audit tables across multi-regional **Google Cloud Storage** buckets.
- Architected database schemas using **ER/Studio** to modernize a core retail banking platform, migrating on-prem infrastructure into unified ledger tables via **Informatica PowerCenter** while validating structural lineage records across the organization inside **Alation** governance catalogs.
- Constructed real-time anti-money laundering tracking workflows by funneling international wire transfers through **Pub/Sub** queues, leveraging complex streaming rules within **Apache Kafka** nodes to flag anomalous account behaviors, and routing alerts to investigation teams.
- Deployed containerized risk assessment microservices onto **Google Kubernetes Engine (GKE)** clusters, utilizing declarative templates managed within **GitHub** and automating deployment validation testing using custom **CI/CD** workflows to guarantee zero-downtime banking system updates.
- Provisioned multi-tier regulatory reporting clusters using **Terraform**, configuring isolated host groups on **Google Compute Engine** to execute complex financial balance operations while strictly enforcing least-privilege administrative access policies through granular cloud-native **IAM** roles.
- Built credit score forecasting pipelines by training optimized **XGBoost** predictive structures on historical loan repayments via **AI Platform (legacy)**, utilizing **Pandas** to compute debt-to-income features before exporting the final model artifacts into secure repositories.
- Implemented an interactive customer support assistant using **Dialogflow** to process routine banking inquiries, integrating custom **Cloud Functions** to fetch real-time checking account balances securely while adhering to corporate **Encryption in Transit** security mandates.
- Maintained continuous platform visibility for high-value wealth management portals by scraping system performance metrics with **Cloud Monitoring**, aggregating microservice traces inside **Cloud Logging**, and diagnosing container resource exhaustion across distributed production node instances.

- Developed an automated document parsing solution for physical loan applications by passing scanned images through **Cloud Vision API**, extracting relevant text fields using **Python** scripts, and storing the validated applicant data securely into **Hive** tables.
- Standardized enterprise business intelligence tracking by creating centralized customer dashboards within **Looker**, using optimized **SQL** queries to isolate quarterly default trends and refreshing underlying reporting metrics daily via automated **IBM DataStage** workflows.
- Trained deep learning asset allocation algorithms inside **Vertex AI**, leveraging historical market data matrices handled by **NumPy** to evaluate portfolio risk profiles before publishing the final tensor scoring models onto highly available cloud endpoints.
- Curated extensive customer sentiment indexes by streaming internal support logs through **Cloud Natural Language API**, utilizing structured data transformations with **dbt** to organize semantic scores, and storing the resulting trends within protected **Delta Lake** files.
- Migrated legacy relational clearinghouse databases into distributed big data architectures, writing high-throughput **SSIS** packages to extract historical check records, transforming the raw strings into tabular formats, and appending the results directly into **HBase** clusters.
- Automated model retraining loops for commercial loan default predictions using **AutoML**, evaluating performance variations across diverse consumer segments with **Matplotlib** visualizations, and managing code version updates through integrated **GitHub Actions** deployment pipelines.
- Orchestrated complex credit card fraud detection routines by executing scheduled **Apache Airflow** jobs, aggregating transaction histories across global **Hadoop** ecosystems, and generating comprehensive risk reporting tables to comply with mandatory international financial auditing regulations.
- Standardized development lifecycles for internal trading APIs by isolating custom **TensorFlow** binary dependencies inside secure **Docker** configurations, subsequently deploying the verified images to production environments governed under structured, collaborative **Agile** development sprint frameworks.
- Maintained a centralized log monitoring framework by indexing core financial microservice traces within an enterprise-scale **ELK Stack**, validating authorization pathways using **Cloud IAM** policies to prevent unauthorized backend database access across internal engineering groups.
- Secured sensitive customer credit bureau reports by implementing hardware-backed **Cloud Key Management Service** keys, ensuring comprehensive **Encryption at Rest** across all persistent disks while utilizing **Hugging Face Transformers** to mask identifiable personal details.
- Formulated an automated loan portfolio classification strategy by training high-performance **LightGBM** models using **Scikit-learn**, converting the raw scoring logic into optimized production routines, and tracking model drift metrics over time using specialized **Seaborn** statistical charts.
- Accelerated customer account onboarding experiences by integrating **Speech-to-Text** components into mobile banking apps, converting voice commands via **Text-to-Speech** confirmation loops, and verifying customer identity by executing secure validation scripts within **Kubernetes** containers.

Tech Stack: Google Compute Engine, Google Kubernetes Engine (GKE), Google Cloud Storage, Cloud IAM, Cloud Functions, Cloud Monitoring, Cloud Logging, Cloud Key Management Service, Vertex AI, AI Platform (legacy), AutoML, Dialogflow, Cloud Vision API, Cloud Natural Language API, Speech-to-Text, Text-to-Speech, PyTorch, Scikit-learn, TensorFlow, Keras, XGBoost, LightGBM, NumPy, Pandas, Matplotlib, Seaborn, BigQuery, Cloud Dataflow, Cloud Dataproc, Pub/Sub, Data Fusion, Dataplex, BigQuery ML, Cloud Composer, Apache Spark, PySpark, Hadoop, Hive, HBase, Delta Lake, Parquet, Apache Kafka, Kafka Connect, Apache NiFi, Apache Airflow, dbt, Informatica PowerCenter, IBM DataStage, SSIS, Alation, ER/Studio, Looker, GitHub, GitHub Actions, Terraform, Docker, Kubernetes, ELK Stack, Python, SQL, Encryption at Rest, Encryption in Transit, IAM, Agile, CI/CD

Walgreens | Deerfield, IL

Big Data Engineer | July 2020 – April 2023

- Captured high-velocity prescription telemetry feeds using **Apache Flume**, steering the unstructured event payloads directly into scalable **HDFS** storage layers before executing daily clinical verification checks, achieving a 40% reduction in total data-ingestion processing latency across target networks.

- Formulated real-time pharmaceutical inventory tracking systems by ingestion of point-of-sale store logs through **Apache Kafka**, transforming message payloads with **PySpark** structured streaming configurations, and persisting the enriched operational metrics into scalable tables within **Amazon DynamoDB**.
- Crafted physical pharmacy schema definitions via **PowerDesigner** to modernize legacy clinical data lakes, mapping complex entity relationships through structured **SDLC** lifecycles to guarantee accurate tracking of patient prescription histories across regional healthcare systems.
- Orchestrated comprehensive patient medication compliance pipelines using **Apache Airflow** DAGs, converting raw nested electronic health records into tabular assets via **Python** automation, and organizing the final clean records inside highly secure **Amazon RDS** instances.
- Standardized enterprise governance architectures by capturing clinical column-level lineage records inside **Apache Atlas**, extracting structural database details via custom **SQL** scripts, and updating data classification rules during collaborative team planning cycles under **Kanban** workflows.
- Generated highly interactive regional drug distribution dashboards within **Amazon QuickSight**, leveraging massive aggregations executed across a centralized **Amazon Redshift** warehouse to deliver immediate operational drug demand insights directly to pharmacy branch managers.
- Provisioned fault-tolerant drug safety analytics infrastructure using **AWS EC2** compute groups, distributing incoming verification application traffic via **Elastic Load Balancing** policies while monitoring global cluster health through automated, multi-region **DataDog** tracing agents.
- Executed high-throughput laboratory result extractions by running distributed **MapReduce** batch processing routines inside self-managed **Hadoop** setups, successfully reducing long-term cloud infrastructure expenditure by 25% while systematically storing the output files across **AWS S3**.
- Managed safe continuous deployment systems for patient-facing mobile portals via **GitLab CI** runners, packaging application binaries inside lightweight **Docker** runtime images and deploying them to verified clusters running production-grade **Kubernetes** orchestrations.
- Streamlined patient prescription refills by capturing incoming checkout event streams with **Amazon Kinesis**, running on-demand data validations through serverless **AWS Lambda** logic, and broadcasting real-time filling status notifications directly back into corporate communications via **Slack**.
- Extracted historical insurance claims out of on-premises transactional databases using **Talend Open Studio**, applying complex deduplication algorithms via **Apache Spark** computing arrays, and staging the final audited financial information within secure **Hive** tables.
- Enforced stringent security boundaries across sensitive pharmacy web systems by setting up granular identity access configurations inside **AWS IAM**, auditing all cross-account cloud transactions using **AWS CloudTrail** logs to maintain continuous compliance with medical standards.
- Configured daily medical catalog synchronization workflows through **AWS Data Pipeline**, processing external pharmacy vendor stock lists with customized **Bash** operational components, and publishing clean inventory updates directly to target teams across **HDFS** structures.
- Maintained continuous platform visibility for high-throughput pharmacy backends by creating real-time alert definitions in **AWS CloudWatch**, streaming infrastructure performance tracking metrics into **Grafana** displays, and managing system troubleshooting tasks using agile **Kanban** frameworks.
- Unified diverse healthcare interaction histories by coordinating multi-stage serverless cataloging tasks using **AWS Glue**, querying underlying clinical schemas via optimized **Hive** structures, and maintaining comprehensive documentation updates over entire system upgrade lifecycles under **SDLC**.
- Built robust clinical research clusters on **AWS EMR** architecture, scaling up parallel execution nodes to process massive electronic medical profiles using **PySpark** analytics, and persisting the finalized statistical records within scalable **Amazon Redshift** data tables.
- Developed stream processing mechanisms using **Apache Beam** to cross-examine prescription pricing rules against dynamic insurer databases, funneling live transaction updates through **Apache Kafka** brokers to prevent billing discrepancies at physical checkout locations.
- Safeguarded sensitive digital health platforms by implementing network isolation policies around container instances on **Kubernetes**, tracking access modifications with **AWS CloudTrail**, and setting up automated failure alerts to report systemic anomalies into engineering channels via **Slack**.
- Standardized internal application build repositories using **GitLab** version controls, launching automated testing scripts through serverless **AWS Lambda** instances, and tracking cloud system memory utilization metrics across **AWS EC2** nodes using centralized **DataDog** monitors.

Tech Stack: AWS EC2, AWS S3, AWS IAM, AWS Lambda, AWS CloudWatch, AWS CloudTrail, Elastic Load Balancing, Amazon Redshift, AWS Glue, AWS EMR, Amazon Kinesis, AWS Data Pipeline, Amazon RDS, Amazon DynamoDB, Apache Spark, PySpark, Hadoop, HDFS, MapReduce, Hive, Apache Beam, Apache Kafka, Apache Flume, Apache Airflow, Talend Open Studio, Apache Atlas, PowerDesigner, Amazon QuickSight, DataDog, Grafana, GitLab, GitLab CI, Docker, Kubernetes, Slack, Python, SQL, Bash, SDLC, Kanban

UPS | Atlanta, GA

Data Engineer | December 2017 – June 2020

- Collected high-velocity telematics data from vehicle fleets using **Apache Flume**, steering the raw streaming information directly into a centralized **HDFS** repository to establish a foundational layer for real-time monitoring of route efficiency metrics.
- Formulated real-time logistics tracking pipelines by capturing fleet coordinate data via **Azure Event Hubs**, transforming the spatial inputs through **Azure Stream Analytics**, and routing the output stream into structured target databases.
- Crafted physical transit data models via **ER/Studio** to design an integrated shipment ledger, executing structured transformations across legacy supply chain platforms using **Informatica** pipelines to standardize multi-regional warehouse inventories.
- Orchestrated daily delivery route processing workloads using **Azure Data Factory** configurations, converting nested transit payloads into relational tables via **Python** scripts, and staging the final audited data directly inside secure **Oracle** tables.
- Standardized enterprise route metadata governance by maintaining structural database schemas within **SQL Server** instances, querying the underlying travel records using optimized **SQL** views during collaborative project milestones under structured **Agile** sprint workflows.
- Generated interactive delivery performance dashboards within **Power BI**, pulling semantic structures out of high-performance semantic models hosted in **Azure Analysis Services** to provide regional transport directors with actionable logistics bottlenecks insight.
- Provisioned scalable computation environments using **Docker** containers to run isolated geospatial parsing scripts, scheduling the container deployment sequences through **Azure Scheduler** jobs to automate midnight maintenance routines.
- Executed high-throughput mileage data extractions by running distributed **MapReduce** batch logic across **Hadoop** frameworks, which reduced long-term reporting storage requirements by 30% while systematically organizing the output files securely.
- Managed continuous integration build systems for scheduling engines using **Jenkins**, committing code updates into centralized repositories with **Git** versioning to ensure stable distribution of backend microservices across target networks.
- Built advanced flight arrival forecasting loops inside cloud environments using **Databricks**, leveraging optimized cluster resources to handle massive flight telemetry profiles before storing the final processed data within **Azure Data Lake Storage Gen1**.
- Extracted historical cargo shipping records from legacy data warehouses using **Teradata** export tools, processing the raw transactional files via automated **Bash** scripts to prepare the ingestion assets for downstream analytical consumption.
- Enforced operational health tracking across transit systems by collecting server metrics with **Splunk**, identifying application exception bottlenecks through centralized dashboard tracking inside **ELK Stack** configurations, and mitigating pipeline performance latency issues.
- Unified diverse vehicle tracking histories by allocating cluster execution memory through **YARN** resources, query-optimizing the heavy tabular transit data with **Hive** scripts, and updating long-term shipping history records across enterprise systems.
- Processed immediate traffic congestion notifications by streaming live sensor inputs through **Apache Storm** networks, which increased on-time delivery arrivals by 15% across major metropolitan zones by dynamically rerouting cargo trucks.

Tech Stack: Azure Data Factory, Azure Data Lake Storage Gen1, Oracle, Databricks, Azure Stream Analytics, Azure Event Hubs, Azure Analysis Services, Python, SQL, Bash, Hadoop, HDFS, YARN, MapReduce, Hive, Apache Storm, Apache Flume, Informatica, SQL Server, Teradata, ER/Studio, Power BI, Git, Jenkins, Docker, Azure Scheduler, Agile, ELK Stack, Splunk

Mastek | Mumbai, India

Data Analyst | December 2015 – October 2017

- Ingested external enterprise application data from corporate **FTP** servers into highly available **Cloud Storage** buckets, leveraging automated **Bash** scripts to orchestrate daily file transfers before transforming raw transactional layouts into clean database records for down-stream analytics engines.
- Constructed scalable product metadata pipelines within **GCP**, utilizing automated extraction workflows inside **Informatica** to migrate legacy operational data into structured tables inside **BigQuery**, which effectively reduced corporate subscription tracking dashboard rendering latencies by over 40%.
- Designed comprehensive semantic layers using **ER/Studio** to support a complex payroll application, implementing multi-tiered **Star Schema** designs to separate operational dimensions from transactional metrics while preserving all architecture revisions within central **SVN** code repositories.
- Orchestrated high-throughput software usage tracking loops by streaming application clickstream telemetry through **Apache Kafka**, applying data cleaning routines across distributed **Hadoop** worker nodes, and appending the sanitized log entries directly into partitioned enterprise **Hive** tables.
- Generated interactive application usage reports inside **Tableau**, establishing direct live database communication channels via secure **ODBC** connections to query high-volume underlying storage tables with highly optimized, analytical window functions written in native **SQL**.
- Executed extensive customer billing behavior studies by processing millions of software subscription profiles using statistical loops written in **R**, pulling structural parameters directly out of **HDFS**, and summarizing quarterly revenue projections within formatted executive **Excel** workbooks.
- Aggregated raw user interaction records by executing parallel **MapReduce** tasks on distributed **Compute Engine** instances, which boosted backend query efficiency by 25% while managing all software release milestones under structured, collaborative **Scrum** project management workflows within the **SDLC**.

Tech Stack: GCP, Compute Engine, Cloud Storage, BigQuery, SQL, R, Bash, FTP, ODBC, Hadoop, HDFS, Hive, MapReduce, Apache Kafka, Informatica, ER/Studio, Tableau, Dimensional Modeling, Star Schema, SVN, SDLC, Scrum, Excel

EDUCATION

University of Mumbai, Mumbai, India
Bachelor of Science in Information Technology

08/11 – 05/15